CDS Coupon, Accrual and Upfront Date Conventions

Date logic in the ISDA CDS Standard Model (C source)
with Bloomberg CDSW verification and a worked 5Y example

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This note sets out how a standardised (post-2009 'Big Bang') CDS contract generates its coupon accrual dates, payment dates, accrued interest and upfront cash-settlement, exactly as coded in the ISDA CDS Standard Model. Every rule is tied to the file and line(s) that implement it. It cross-checks the OpenGamma paper and Bloomberg CDSW against the same code, and ends with a fully worked 5Y example (Appendix A). File paths are relative to `lib/src/`. Dates throughout include their weekday.

Summary at a glance

- **Payment dates** are the IMM 20th, Following-adjusted to the next good business day.
- **Accrual dates** are Following-adjusted IMM dates. The *first accrual start* is the prior IMM used as-is (unadjusted); the *final accrual end* is the maturity IMM **plus one calendar day**.
- **Accrued interest** = N · coupon · ACT/360 from the current period's accrual start to the step-in date (valuation date + 1 calendar day), undiscounted (feeleg.c:587).
- **Value date**: the premium and protection legs are discounted to the value date supplied; the accrued is undiscounted (Section 5).
- **Quotation**: new deals are quoted as a cash settlement (value date = T+3 business days); existing deals are quoted at mark-to-market (value date = today) — Section 6.
- **Bloomberg** matches the ISDA model on every date; the final coupon differs by the calendar +1 day when the maturity is a good business day (Section 8).

1. Model inputs and the key dates

The model takes the trade, step-in, value (cash-settle), start and end dates as *inputs* and applies fixed rules to build the schedule. The calling layer sets the standardised values below.

Input	Meaning	Standard value
today (T)	Trade date / curve base date.	Trade date
stepinDate	Protection effective date; the date the issuer becomes risky and the accrued cut-off.	T + 1 calendar day
valueDate (settleDate)	The date the PV is discounted to; the upfront cash settles here.	T + 3 business days
startDate	Accrual begin / prior coupon date; the first period accrues from here.	Prior IMM date before T (unadjusted)
endDate	Maturity; protection and accrual end here.	Unadjusted IMM date (20 Mar/Jun/Sep/Dec)

In `JpmcdsCdsPrice` the value date equals the settle date: `valueDate = settleDate` (cds.c:324). The fee leg is built from `startDate` and `endDate` by `JpmcdsCdsFeeLegMake` (cds.c:117–229).

2. Coupon accrual and payment dates

The raw period boundaries (the IMM dates) are generated **unadjusted** by `JpmcdsDateListMakeRegular` (cxdatelist.c:329) as pure interval arithmetic. Bad-day adjustment is then applied in the schedule loop of `JpmcdsCdsFeeLegMake` (cds.c:181–214):

```
prevDate    = dl->fArray[0];
prevDateAdj = prevDate;             /* first date is NOT bad-day adjusted  (cds.c:182) */
for (i = 0; i < fl->nbDates; ++i) {
  JpmcdsBusinessDay(nextDate, badDayConv, calendar, &nextDateAdj); /* cds.c:189 */
  fl->accStartDates[i] = prevDateAdj;   /* cds.c:192 */
  fl->accEndDates[i]   = nextDateAdj;   /* cds.c:193 */
  fl->payDates[i]      = nextDateAdj;   /* cds.c:194 */
}
if (protectStart) {                             /* TRUE for standard CDS */
  fl->accEndDates[nbDates-1] = prevDate + 1;   /* unadjusted IMM + 1 CALENDAR day
(cds.c:207) */
  fl->obsStartOfDay = TRUE;                     /* cds.c:208 */
}
```

The rules resolve to:

- **Interim coupons.** Accrual start, accrual end and payment date are the IMM dates adjusted to the next good business day (Following) — cds.c:189, 192–194.
- **First coupon.** The accrual start is the prior IMM date used as-is, unadjusted (cds.c:182). The accrual end and payment date are the next IMM, Following-adjusted.
- **Final coupon.** The accrual end is the unadjusted maturity IMM date plus one calendar day — literally `prevDate + 1` (cds.c:207). Because `TDate` is an integer day count, this adds exactly one **calendar** day (for example Fri 20-Jun → Sat 21-Jun). Start-of-day observation (`obsStartOfDay = TRUE`, cds.c:208) pairs with it so protection and accrual include the maturity date. The final payment date is the maturity IMM, Following-adjusted.

Period	Accrual start	Accrual end	Payment date
Initial coupon	Prior IMM, unadjusted	Next IMM, Following-adjusted	= accrual end
Interim coupons	Prior IMM, Following-adjusted	Next IMM, Following-adjusted	= accrual end
Final coupon	Prior IMM, Following-adjusted	Maturity IMM + 1 calendar day	Maturity IMM, Following-adjusted

> **Common pitfalls.** Leaving the interim accrual dates unadjusted (the model adjusts them Following); bad-day-adjusting the first accrual start or the final accrual end (the first start is unadjusted, the final end is a raw calendar +1); and treating the first accrual start as the step-in / T+1 date (it is the prior IMM, while T+1 serves only as the accrued cut-off).

The payment date discounts the surviving coupon; the period day-count uses accrual start to accrual end (`FeePaymentPVWithTimeLine`, feeleg.c:248, 252–254).

3. The upfront payment

The upfront (points-up-front, clean price) is the clean PV of the CDS. `JpmcdsCdsoneUpfrontCharge` (cdsone.c:53–129) bootstraps a flat credit curve from the quoted spread and returns the price from `JpmcdsCdsPrice` (cds.c:292) with `isPriceClean = TRUE`:

- Upfront = Contingent-leg PV − clean Fee-leg PV (cds.c:361).
- Both legs are discounted to the value date — the cash-settle date, T+3 (Sections 5 and 6).
- The accrued-interest rebate is subtracted to make the price clean (Section 4).

So the upfront uses the accrual dates of Section 2, protection from `startDate` to `endDate`, and a single cash-settlement date (the value date, T+3) to which the legs are discounted.

4. Accrued interest — clean and dirty

Accrued interest is the premium built up from the start of the current coupon period to the step-in date. Clean pricing subtracts it from the premium-leg PV; dirty pricing leaves it in. The switch is `payAccruedAtStart` / `cleanPrice` in `JpmcdsFeeLegPV` (feeleg.c:162–171):

```
if (payAccruedAtStart) {            /* clean price (feeleg.c:162) */
    FeeLegAI(fl, stepinDate, &ai);    /* accrued to step-in date (feeleg.c:165) */
    *pv -= ai;                        /* subtract, UNDISCOUNTED (feeleg.c:170) */
}
```

The accrued amount is computed in `FeeLegAI` (feeleg.c:551–600):

```
JpmcdsDayCountFraction(fl->accStartDates[lo], today, fl->dcc, &accrual); /* feeleg.c:587 */
accrual *= fl->couponRate * fl->notional;                               /* feeleg.c:590 */
```

Accrued-interest detail	Value
Accrual start	Start of the coupon period containing the step-in date: the prior Following-adjusted IMM for interim periods, or the unadjusted contract start in the first period (feeleg.c:587).
Cut-off (end)	The step-in date = valuation date + 1 calendar day, supplied as 'today' to FeeLegAI (feeleg.c:165).
Day count	Accrual DCC, almost always ACT/360 (feeleg.c:587).
Discounting	None: the accrued is a raw cash amount (feeleg.c:170), independent of the value date. It is subtracted from a leg PV that is already at the value date, so it has an implicit discount factor of 1.

One accrued calculation across the life of the trade

The accrued is a single calculation, evaluated wherever the valuation sits: from the current period's accrual start to the step-in date (the valuation date + 1 calendar day), on ACT/360, undiscounted. The accrual start is the current period's start — the prior Following-adjusted IMM, or the unadjusted contract start in the first period. Two observations follow:

- **On the trade date**, the step-in is T+1 and the accrued is the rebate the protection buyer receives within the upfront cash.

- **For an existing deal**, the step-in is the valuation date + 1, and the same accrued is the amount that converts that day's dirty PV into the clean PV. It grows day by day and resets to zero at each accrual-start date.

Because the accrued is undiscounted, the choice of value date (Section 6) does not change it. What advances the accrued over the life of the trade is the step-in date, not the value date.

> **Common pitfall.** Using the value / cash-settle date as the accrued cut-off; the cut-off is the step-in date (valuation date + 1 calendar day), which is earlier by the settlement lag.

5. Discounting and the value date

Every price is a present value as of one chosen date: the value date. The engine prices each leg's cashflows to `today` (the trade / curve base date) and then applies the discount factor from `today` to the value date — `pv = myPv / JpmcdsForwardZeroPrice(discCurve, today, valueDate)` (feeleg.c:158–160; contingentleg.c:146–150). The value date equals the `settleDate` input (cds.c:324).

This discount factor scales the **premium and protection legs**. The accrued interest is a raw cash amount, added undiscounted (feeleg.c:170), so it sits at the value date with a factor of 1. Choosing the value date therefore rescales the two legs and leaves the accrued unchanged.

When the value date is later than a cashflow's payment date, the factor exceeds 1 — a **growth factor** rather than a discount factor. In the worked example (Appendix A) the value date is the settlement date and the first coupon pays one day earlier, so its factor is 1.000107.

6. Quotation conventions: cash settlement and mark-to-market

The same engine produces two standard numbers. They use the same schedule and the same accrued, and differ only in the value date:

- **Cash settlement** — the cash that physically changes hands. Cash settles three business days after the trade (or unwind) date, so the value date is that cash-settle date (T+3 business days), and both legs are discounted to it. **New deals are quoted this way** (the upfront).
- **Mark-to-market** — the market value for P&L and risk. The value date is the valuation date itself, so the factor from today to today is 1 and the PV is reported as of today. **Existing deals are quoted this way**.

So a new deal on its trade date is quoted as the cash settlement (value date = T+3); an existing deal is quoted at mark-to-market on the valuation date (value date = today). The two numbers differ only by the discount (or growth) factor between today and the settlement date applied to the legs; the accrued is identical in both. A later cash event — an unwind, novation or assignment of an existing deal — is itself a cash settlement, so its value date is that event's own T+3.

Worked examples (the 5Y deal from Appendix A)

1. New deal — cash-settlement quote. Trade date **Thu 18-Jun-2026**. The upfront settles three business days later, so the value date is **Tue 23-Jun-2026**. Both legs are discounted to Tue 23-Jun-2026, and the result is the upfront the buyer pays that day (≈ $635,379).

2. Existing deal — mark-to-market quote. Valuation date **Mon 20-Jul-2026**. The value date is the valuation date itself, **Mon 20-Jul-2026**. The discount factor from Mon 20-Jul-2026 to Mon 20-Jul-2026 is 1, so the PV is reported as of that day. No three-day shift is applied.

3. Closing an existing deal — cash settlement again. Unwind agreed on **Mon 20-Jul-2026**. The unwind cash settles three business days later, so the value date is **Thu 23-Jul-2026**, and both legs are discounted to Thu 23-Jul-2026 to give the unwind cash amount.

Quotation	Used for	Value date (PV expressed as of)
Cash settlement (upfront)	New deals	Cash-settle date = T + 3 business days
Mark-to-market	Existing deals	The valuation date (today); factor = 1
Unwind / novation (cash)	Closing an existing deal	Valuation date + 3 business days

The three-day shift therefore belongs to cash settlement, not to the trade date. A new-deal quote is a cash-settlement figure, so its value date is T+3; an existing-deal quote is a mark-to-market, so its value date is today. The shift reappears only when an existing deal is closed for cash. Bloomberg CDSW exposes both controls — a *Valuation Date* and a *Cash Settled On* date — matching the two conventions.

7. Agreement with OpenGamma

The OpenGamma paper (Sections 2.1 and 4.3) agrees with the code on every material point:

- Step-in = T+1, cash-settle ≈ T+3, start = prior IMM, maturity = unadjusted IMM, payment dates = IMM adjusted to the next good business day. **Matches.**
- “Apart from the final accrual end date, [accrual start/end] are adjusted IMM dates” using Following. **Matches** cds.c:189–194.
- Clean RPV01 = dirty RPV01 − N·DCC(s_1, t_e), s_1 the accrual start immediately before step-in, t_e the step-in date (their eq. 21). **Matches** FeeLegAI (feeleg.c:587), with the step-in date as the cut-off.
- Market value discounts to the valuation date and cash settlement to the cash-settle date (their Section 4.3.2). **Matches** the value-date division at feeleg.c:160 and contingentleg.c:150.

The paper describes the final accrual end as the “(unadjusted) IMM date.” The code stores it as the unadjusted IMM date **plus one calendar day** with start-of-day observation (cds.c:207–208). As Section 8 shows, this one-day point is where Bloomberg also diverges; the code is the binding specification.

8. Bloomberg CDSW verification

Trades validated

Two standard sovereign CDS, taken from the CDSW screens, were used for the reconciliation:

Term	Value (as shown on CDSW)
Reference entity	Federative Republic of Brazil — sovereign, Senior tier, 2014 transaction type, restructuring CR14
Side	Buy protection
Notional	USD 10,000,000
Coupon	100 bp
Day count / frequency	ACT/360, quarterly
Recovery / pay-accrued	25% / pay accrued on default = yes
Trade date	Thu 18-Jun-2026
Accrual begin (prior IMM)	Fri 20-Mar-2026
Effective (step-in)	Fri 19-Jun-2026 (T+1)
Cash settled / calculated	Wed 24-Jun-2026 / Tue 23-Jun-2026 (per screen; ≈ T+3)
Tenors reconciled	1Y (maturity Sun 20-Jun-2027) and 3Y (maturity Wed 20-Jun-2029)

Method

The official ISDA C library in this repository was compiled and the fee-leg schedule generated by `JpmcdsCdsFeeLegFlows` (cds.c:463) for both tenors, then reconciled against the CDSW cashflow screens. Bloomberg's "Date" column is the **payment date**; the values are exactly the Following-adjusted IMM dates.

Results

Every payment date and every accrual period matches the official ISDA model to the cent. The final coupon agrees too, except on the calendar +1 day when the maturity IMM is a good business day:

- Payment dates: **all match** (Following-adjusted IMM dates).
- First and interim coupons: **match exactly**, including the 94-day first period and the 92-day longer periods.

3Y contract, maturity Wed 20-Jun-2029 (a good business day):

Final coupon	ISDA Standard Model (official code)	Bloomberg CDSW
Payment date	Wed 20-Jun-2029	Wed 20-Jun-2029
Accrual end	Thu 21-Jun-2029 (maturity + 1 calendar day)	Wed 20-Jun-2029 (maturity)
Accrual days (ACT/360)	93	92

Final coupon	ISDA Standard Model (official code)	Bloomberg CDSW
Coupon amount	$25,833.33	$25,555.56
Difference	—	**1 day · $277.78 on $10mm**

The ISDA model's final-period extra day comes from `accEndDates[last] = prevDate + 1` (cds.c:207) with `obsStartOfDay = TRUE` (cds.c:208), a calendar day. The official code accrues the last premium to maturity + 1 calendar day; Bloomberg accrues it to the maturity IMM date.

The two agree when the maturity IMM falls on a weekend or holiday. For the 1Y contract maturing **Sun 20-Jun-2027**, Following adjustment moves the payment to **Mon 21-Jun-2027**, which equals maturity + 1, so both give 91 days. Because the 20th lands on a weekend roughly two years in seven, the difference is visible on the majority of standard maturities. The OpenGamma prose ("unadjusted IMM date", 92 days here) aligns with Bloomberg; the official code adds the extra calendar day (93 days).

Verdict

Bloomberg's payment dates and all first and interim accrual dates match the ISDA model exactly. The final-coupon accrual runs to the maturity IMM date, one calendar day short of the model's maturity + 1, giving a one-day difference on the last premium (here $277.78, ≈0.0028% of notional) when the maturity is a good business day — small in PV terms, and a clean, reproducible point to keep in mind when reconciling to the official model.

Source-code reference map

What	File	Line(s)
Fee-leg / schedule builder	cds.c	117–229
Raw unadjusted IMM schedule	cxdatelist.c	329 (DateListMakeRegular)
First accrual start unadjusted	cds.c	182
Following adjustment of period dates	cds.c	189
accStart / accEnd / payDate assignment	cds.c	192 / 193 / 194
Final accrual end = IMM + 1 calendar day, obs start-of-day	cds.c	207–208
valueDate = settleDate	cds.c	324
CDS price = ContingentPV − FeePV	cds.c	361
Fee-leg PV; discount to value date	feeleg.c	158, 160
Contingent-leg PV; discount to value date	contingentleg.c	146–150
Clean: subtract accrued, undiscounted	feeleg.c	162–171
Accrued amount, start→step-in, ACT/360	feeleg.c	551–600 (587, 590)
Surviving coupon discounted to pay date	feeleg.c	248, 252–254
Upfront charge entry point	cdsone.c	53–129
Nominal fee cashflows (used for verification)	cds.c	463 (JpmcdsCdsFeeLegFlows)

Appendix A — Worked example: 5Y CDS

A cash-settlement (new-deal) quotation: value date = T+3. All dates and cashflows are produced by the official ISDA C library; discount factors and survival probabilities use a flat 4% (ACT/365F) discount curve and a credit curve bootstrapped from a 250 bp par spread at 25% recovery. These curve levels are illustrative, while the dates and day-counts are exact.

Contract terms and key dates

Term	Value	Note
Side / notional	Buy protection · USD 10,000,000	—
Coupon / day count	100 bp · ACT/360 · quarterly	Standard coupon
Recovery	25%	Protection leg
Trade date (T)	Thu 18-Jun-2026	Curve base date
Effective / step-in	Fri 19-Jun-2026	T + 1 calendar day
Value / cash-settle date	Tue 23-Jun-2026	T + 3 business days
Accrual begin (startDate)	Fri 20-Mar-2026	Prior IMM, unadjusted
Maturity (endDate)	Fri 20-Jun-2031	5Y IMM, unadjusted

Premium-leg schedule and cashflows

Raw IMM	Accrual start	Accrual end	Payment date	Days	Coupon (USD)	Disc factor	Survival	Coupon PV (USD)
Sat 20-Jun-2026	Fri 20-Mar-2026	Mon 22-Jun-2026	Mon 22-Jun-2026	94	26,111.11	1.000107	0.999724	26,106.70
Sun 20-Sep-2026	Mon 22-Jun-2026	Mon 21-Sep-2026	Mon 21-Sep-2026	91	25,277.78	0.990376	0.991377	24,818.62
Sun 20-Dec-2026	Mon 21-Sep-2026	Mon 21-Dec-2026	Mon 21-Dec-2026	91	25,277.78	0.980739	0.983099	24,371.91
Sat 20-Mar-2027	Mon 21-Dec-2026	Mon 22-Mar-2027	Mon 22-Mar-2027	91	25,277.78	0.971195	0.974891	23,933.25
Sun 20-Jun-2027	Mon 22-Mar-2027	Mon 21-Jun-2027	Mon 21-Jun-2027	91	25,277.78	0.961745	0.966751	23,502.48
Mon 20-Sep-2027	Mon 21-Jun-2027	Mon 20-Sep-2027	Mon 20-Sep-2027	91	25,277.78	0.952387	0.958680	23,079.46
Mon 20-Dec-2027	Mon 20-Sep-2027	Mon 20-Dec-2027	Mon 20-Dec-2027	91	25,277.78	0.943119	0.950675	22,664.06
Mon 20-Mar-2028	Mon 20-Dec-2027	Mon 20-Mar-2028	Mon 20-Mar-2028	91	25,277.78	0.933942	0.942738	22,256.13
Tue 20-Jun-2028	Mon 20-Mar-2028	Tue 20-Jun-2028	Tue 20-Jun-2028	92	25,555.56	0.924755	0.934780	22,091.31
Wed 20-Sep-2028	Tue 20-Jun-2028	Wed 20-Sep-2028	Wed 20-Sep-2028	92	25,555.56	0.915658	0.926890	21,689.37
Wed 20-Dec-2028	Wed 20-Sep-2028	Wed 20-Dec-2028	Wed 20-Dec-2028	91	25,277.78	0.906748	0.919151	21,067.47
Tue 20-Mar-2029	Wed 20-Dec-2028	Tue 20-Mar-2029	Tue 20-Mar-2029	90	25,000.00	0.898021	0.911561	20,465.02
Wed 20-Jun-2029	Tue 20-Mar-2029	Wed 20-Jun-2029	Wed 20-Jun-2029	92	25,555.56	0.889187	0.903867	20,539.17
Thu 20-Sep-2029	Wed 20-Jun-2029	Thu 20-Sep-2029	Thu 20-Sep-2029	92	25,555.56	0.880440	0.896237	20,165.47
Thu 20-Dec-2029	Thu 20-Sep-2029	Thu 20-Dec-2029	Thu 20-Dec-2029	91	25,277.78	0.871873	0.888754	19,587.27

Raw IMM	Accrual start	Accrual end	Payment date	Days	Coupon (USD)	Disc factor	Survival	Coupon PV (USD)
Wed 20-Mar-2030	Thu 20-Dec-2029	Wed 20-Mar-2030	Wed 20-Mar-2030	90	25,000.00	0.863482	0.881415	19,027.15
Thu 20-Jun-2030	Wed 20-Mar-2030	Thu 20-Jun-2030	Thu 20-Jun-2030	92	25,555.56	0.854988	0.873975	19,096.09
Fri 20-Sep-2030	Thu 20-Jun-2030	Fri 20-Sep-2030	Fri 20-Sep-2030	92	25,555.56	0.846577	0.866598	18,748.64
Fri 20-Dec-2030	Fri 20-Sep-2030	Fri 20-Dec-2030	Fri 20-Dec-2030	91	25,277.78	0.838340	0.859363	18,211.06
Thu 20-Mar-2031	Fri 20-Dec-2030	Thu 20-Mar-2031	Thu 20-Mar-2031	90	25,000.00	0.830271	0.852266	17,690.30
Fri 20-Jun-2031	Thu 20-Mar-2031	Sat 21-Jun-2031	Fri 20-Jun-2031	93	25,833.33	0.822104	0.845072	17,947.38

Disc factor is to the value date (Tue 23-Jun-2026); the Mon 22-Jun-2026 coupon shows a factor just above 1 (a growth factor) as it falls one day before the value date. Survival is observed start-of-day (cds.c:208). Coupon PV = amount × disc factor × survival; the column sums to USD 447,058 (coupons only), and the model's premium-leg PV adds the accrual-on-default term.

Valuation (value date Tue 23-Jun-2026)

Quantity	Amount (USD)	Note
Premium leg PV — dirty	448,863.70	Coupons (447,058) + accrual-on-default (1,805)
Accrued (91 days, Fri 20-Mar→Fri 19-Jun)	25,277.78	N · 1% · 91/360, undiscounted
Premium leg PV — clean	423,585.92	= dirty − accrued
Protection leg PV	1,058,964.80	N · (1−R) · ∫ Z dQ
Clean upfront — buyer pays	**635,378.88**	= protection − clean premium (≈ 6.354 points)
Dirty PV	610,101.10	= protection − dirty premium

The final coupon (payment Fri 20-Jun-2031) accrues to Sat 21-Jun-2031 — the maturity IMM plus one calendar day — giving 93 days and USD 25,833.33, the largest regular coupon, and showing the ISDA extra-day rule with the accrual end landing on a weekend.